MMA3001 – Project Report

Engineering Problem

The engineering problem addressed in this project was to develop a computational method capable of isolating the main structure of a tree from RGB and depth images. This was achieved using winter observations, where the absence of leaves makes the branches more visible.

This has applications to surveillance environments such as automatic flying or driving systems where it needs to be able to identify specific environmental features.
The aim of this project was to develop and evaluate a two-dimensional tree isolation algorithm using image processing techniques, providing a potential foundation for further tree reconstruction.

AI-use Reflection

What AI Was Used

Throughout this project, OpenAI's ChatGPT was used to assist with the development of Python functions, troubleshooting errors and improving code documentation. One limitation of using AI was that its suggestions were not always appropriate for the specific dataset or computational problem. Therefore, AI-generated code needed to be tested and evaluated rather than assumed to be correct.

How AI Was Used

AI was mainly used to help identify errors in the code, implement concepts as Python functions and identify different libraries that could be used to improve the algorithm.
One example was during the development of the tree isolation functions, where AI assisted with structuring the code and troubleshooting errors. However, the effectiveness of the suggested methods still needed to be assessed by visually inspecting the resulting masks and comparing them with the expected tree structures.
An example where AI suggested something that I then rejected was when AI tried to progress the project towards a construction of 3D trees rather than focusing on the 2D reconstruction.
AI was also used to improve the documentation of the final Python files, including adding function descriptions, parameter explanations and comments. This made the code easier to understand without changing the underlying computational methods.
Additionally, AI was used to edit and improve writing flow for this report, but all writing and changes were validated. AI rewrote some sections, which I then revised and checked.
AI also generated with overview the README file based off all the code and this report.

Statement of Human Oversight and Responsibility

AI-generated code was evaluated by running it and examining the results against the expected outputs. The final computational approach was developed through experimentation with the dataset, visual inspection and quantitative evaluation.

All final decisions regarding the computational approach, parameter selection, validation and interpretation of results remained my responsibility. I take full responsibility for the final code, results and conclusions presented in this project.

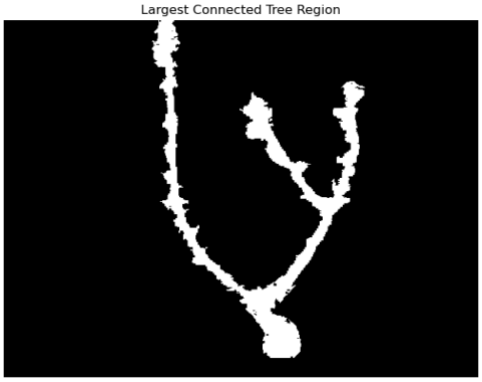
Inputs, Outputs and Data

Inputs

There are 2 main inputs to this algorithm. The first is the depth information provided in a png file where the colour represents the depth from the camera. The maximum depth was 145 while the minimum was 0. The second input was the winter RGB photos which were also provided as a png file. Both inputs had a shape of (480, 640, 3) which mean that the height was 480 pixels, the width was 640 pixels and there was 3 colour streams (RGB).

Outputs

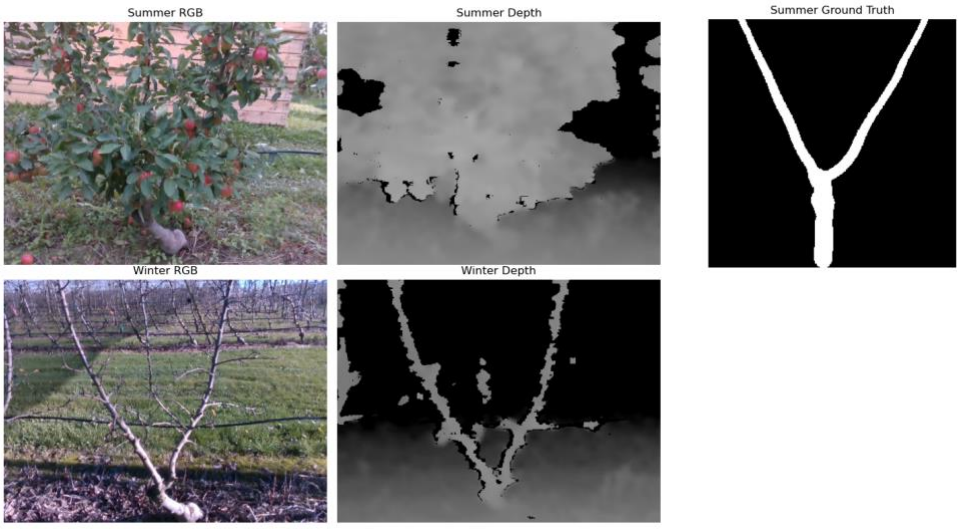
The output of the algorithm is a mask of size 480 X 640 with Boolean values populating it. This can be displayed as an image of the body of the tree.



Data

The data set used in this project contains images of trees captured in both winter and summer. The summer trees had depth images, RGB images as well as ground- truth (GT) information, whereas the winter observations only had the depth and RGB images.

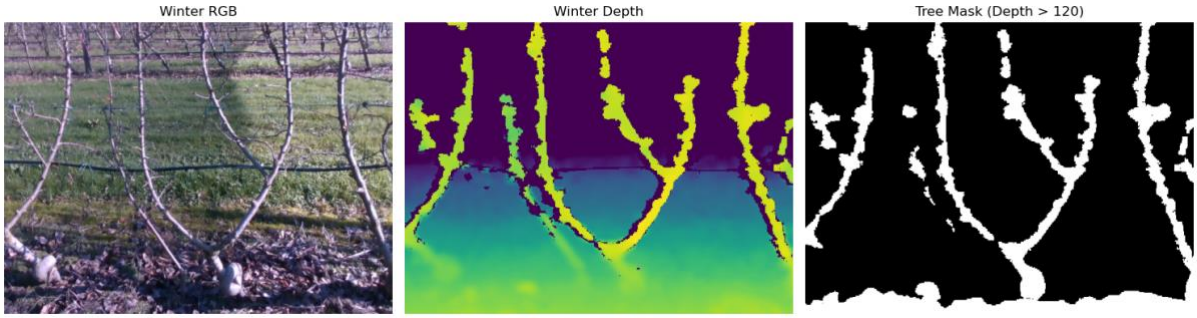
The GT images however were of a different shape ((256, 256)) then the other images. It was also found that the GT images were taken from a different orientation than the winter observations.



Computational Approach

Depth Thresholding

The first method used to isolate the tree was the depth png images. The depth png's had a range of [0,145]. The idea was to select a threshold value and remove all information beyond that. A threshold was selected through visual inspection to be 120. So, a mask was created that only included values of a depth greater than 120.



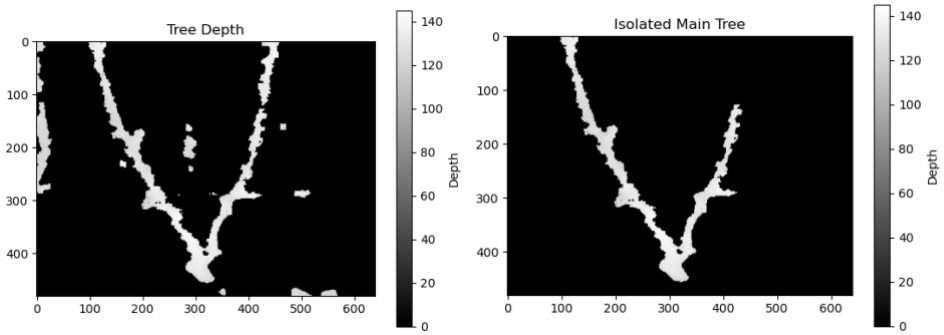
Connected Component Analysis

The depth thresholding was effective at identifying parts of the tree, however it also included other trees and objects that were at a similar depth.

To overcome this, I implemented a connected component analysis to identify the main tree body. This method works by examining each pixel and then determining which other pixels were connected to it. An 8-neighbour approach was used, meaning that pixels could be connected horizontally, vertically or diagonally.

Each connected region was then grouped, and the total number of pixels in each region was calculated. The largest connected region was selected as the main tree, while the remaining regions were removed.

This approach assumed that the tree being investigated would form the largest connected region in the image. Although this worked for many observations, it could fail if another object or neighbouring tree formed a larger connected region.



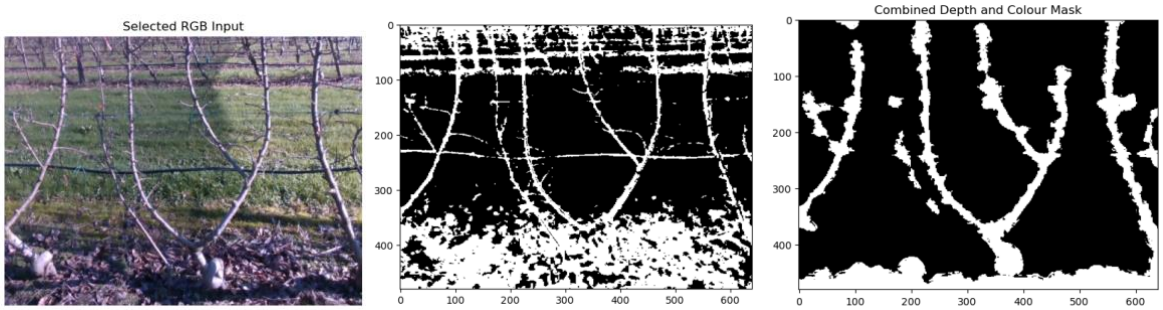
Colour Segmentation

Even after implementing the connected component analysis there was still a lot of noise in the image and if the depth thresholding caused a small gap in the tree some of the tree will be removed.

To fix this I decided to implement another input that was in the data set. Before this I had only used the depth image, so I decided to use the RGB image as well. I had used a similar approach in a previous personal project to isolate what I wanted from an image.

I started by taking some sample data from the tree image to be able to identify any pattern with the colours that I could use to isolate the tree. It was found through a visual inspection that a good isolation was when you take the integer value of the blue colour and subtract the green colour it will isolate the bark colour when the difference is greater than 18.

From this alone you could visually see the tree, however as the ground was a similar colour and there was lots of noise with this method it was combined with the depth mask. To better isolate the tree.



Ground removal

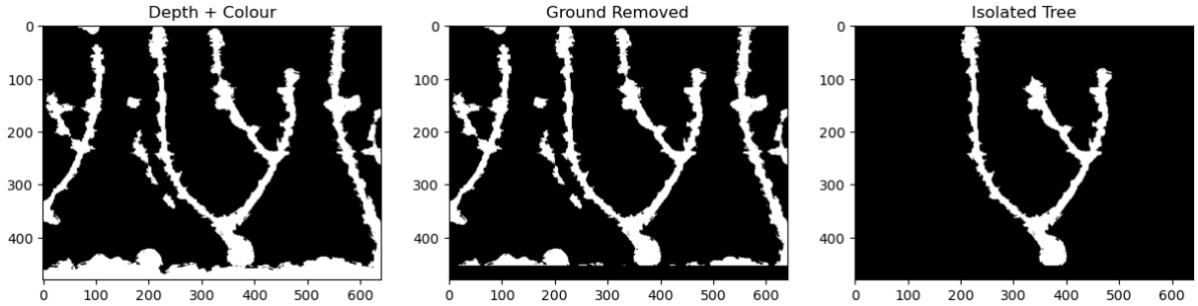
All of the methods that had been implemented so far still had one main issue. The ground was not always being removed from the image because it could have a similar depth and colour to the tree. This could cause neighbouring trees or other background objects to become connected to the main tree during connected component analysis.

To overcome this, I identified that when the ground was included in the segmentation, it often extended across a large proportion of the image width. Therefore, I implemented a method that removed horizontal rows when the proportion of foreground pixels exceeded a specified threshold.

Initially, this threshold was set to 0.7, meaning that a row would be removed if more than 70% of its pixels were selected by the previous criteria. This value was selected through visual experimentation to remove the ground while retaining as much of the tree as possible.

The ground threshold was later included in the parameter optimisation process, where a value of 0.5 was selected as the best-performing value among those tested.

After removing the ground and applying connected component analysis, the algorithm was able to isolate the main tree more effectively from the surrounding background.



Validation Method and Results

Ground Truth Alignment

After implementing the computational approach, I realised that the summer ground-truth images were taken from different angles and distances compared with the winter observations. This created a significant challenge when trying to validate the results, as the ground truth could not be directly compared with the isolated tree masks.

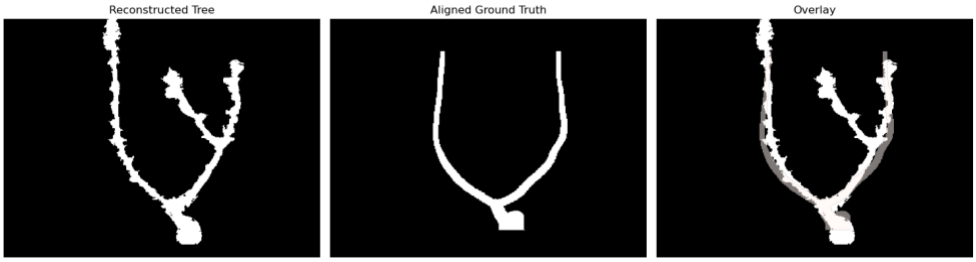
The first step was to resize ethe ground-truth images to that they were approximately the same scale as the isolated trees. Through visual inspection, a scaling factor of 0.75 was selected. Although a constant scaling factor could be used, there was no consistent translations that could work to overlap the trees.

To overcome this, I developed an algorithm that automatically estimated a common point in both images. As the trees generally had a V-shaped structure where the main branches separated, I decided to use this junction as the reference point for alignment.

The algorithm first resizes the ground-truth image using nearest-neighbour interpolation. It then searches for the V-junction in both the resized ground truth and the isolated tree mask. To identify the junction, the algorithm examines horizontal rows, beginning below the upper 15% of the detected tree's vertical extent. It identifies rows containing two sufficiently wide foreground regions separated by a gap.

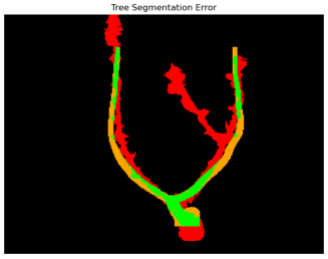
Consecutive rows containing a suitable gap are grouped, and the lowest qualifying gap is used to estimate the V-junction. The ground-truth image is then translated horizontally and vertically so that its estimated junction coincides with the junction in the isolated tree.

This method allowed the alignment process to be automated across the dataset. However, the alignment remained approximate because it used a fixed scaling factor and a single reference point. It could not fully correct differences in camera perspective, rotation or the visible structure of the trees.



Evaluation metrics

4 evaluation metrics were used: intersection over union (IoU), dice coefficient, precision and recall. They are all very similar metrics however they all place emphasis on slightly different aspects.



Intersection over Union (IoU)

How this evaluation metric works is it takes the overlap of the tree and the GT and divides it by the total area that they both cover. This is particularly useful as it penalises both missed pixels as well as incorrectly included pixels.

Dice Coefficient

The dice coefficient works in a very similar way to the intersection over union however it places a greater relative weight to how closely the isolated tree matches the ground truth.

Precision

Precision is the proportion of pixels classified as tree pixels that actually belong to the ground-truth tree mask. High precision indicates at it is good at isolating the tree, whereas low precision indicates that there is too much of the background in the isolation.

Recall

Recall measures how many of the actual tree pixels your algorithm successfully identified. High recall indicates that your algorithm captures most of the tree, including smaller branches. Low recall suggests that parts of the tree are being missed, potentially because of depth or colour thresholding.

Validation results

The tree isolation algorithm was evaluated across the full dataset of 562 trees using the original parameters selected during development. Of these, 422 trees were successfully evaluated against the aligned ground truth, while 140 could not be successfully evaluated. A successful evaluation meant that the segmentation and alignment process completed and produced numerical metrics, rather than guaranteeing that the ground-truth alignment was accurate.

	Mean	Minimum	Maximum	Standard Deviation
IoU	0.2019	0.0076	0.4772	0.0957
Dice	0.3250	0.0151	0.6461	0.1382
Precision	0.2695	0.0091	0.7056	0.1355
Recall	0.4495	0.0450	0.8017	0.1608

The average Intersection over Union (IoU) was 0.2019, while the average Dice coefficient was 0.3250. These results indicate relatively low overlap between the isolated tree masks and the aligned ground truth. There was also considerable variation between different trees, with IoU values ranging from 0.0076 to 0.4772 and Dice coefficients ranging from 0.0151 to 0.6461.

The average precision was 0.2695, compared with an average recall of 0.4495. This suggests that the algorithm was more successful at identifying some of the pixels belonging to the tree than at excluding unwanted pixels. However, these results are also influenced by the accuracy of the ground-truth alignment.

The standard deviations of the IoU and Dice coefficient were 0.0957 and 0.1382 respectively, indicating that the measured performance varied considerably between different trees.

Overall, the results show that the algorithm was capable of identifying tree structures, although its measured agreement with the aligned ground truth was relatively low. The results should therefore be considered alongside the limitations of the validation method.

Validation limitations

The main limitation of the validation process was that the summer ground truth and winter observations were captured from different distances and angles. This meant that even when the tree was successfully isolated, the predicted mask and ground truth did not necessarily overlap accurately.

Because the evaluation metrics compare the images pixel by pixel, differences in camera viewpoint, scale and visible branch structure could significantly reduce the calculated scores independently of the segmentation algorithm's performance.

Another limitation was that only one reference point was used to align the images. Although the V-junction provided a useful feature for translation, it could not account for rotation or perspective differences. Identifying additional corresponding points could allow more accurate transformations to be applied.

Furthermore, 140 trees did not produce successful evaluations. These unsuccessful cases were excluded from the calculated summary statistics, meaning the reported results represent only the trees for which the evaluation pipeline completed.

Consequently, the evaluation results reflect both the accuracy of the tree isolation algorithm and the accuracy of the ground-truth alignment, rather than providing a completely independent measurement of segmentation performance.

Comparison and Optimisation Work

Intersection over Union (IoU) was selected as the main evaluation metric for optimising the tree isolation algorithm because it considers both incorrectly included pixels and missed tree pixels.

Four parameters were investigated: the depth threshold, colour threshold, dilation iterations and ground threshold. Each parameter was varied individually while the remaining parameters were held at their original values. This allowed the effect of each parameter on the measured segmentation performance to be investigated.

Initially, the parameters were tested on a single tree to visually examine how changing their values affected the segmentation. The investigation was then extended to the full dataset of 562 trees, where the mean IoU was calculated for each tested parameter value.

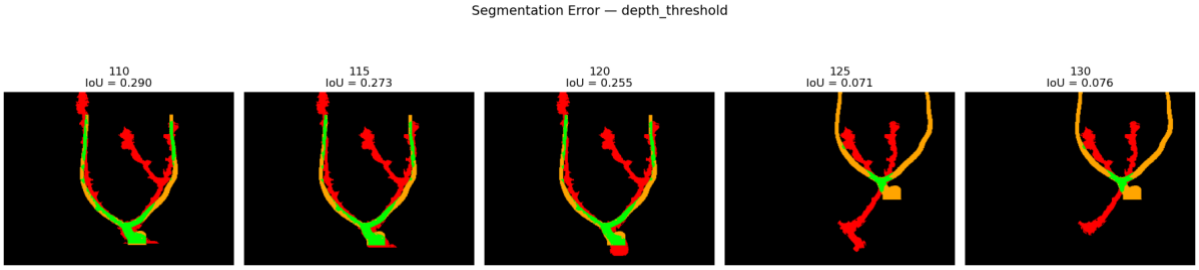
The values producing the highest mean IoU were then combined to create an optimised configuration. Finally, the original and optimised configurations were compared using a smaller sample of trees to determine whether the changes improved the measured segmentation performance.

Optimisation For Single Tree

Depth Threshold

The depth threshold controls which pixels are initially identified as potentially belonging to the tree based on their depth values.

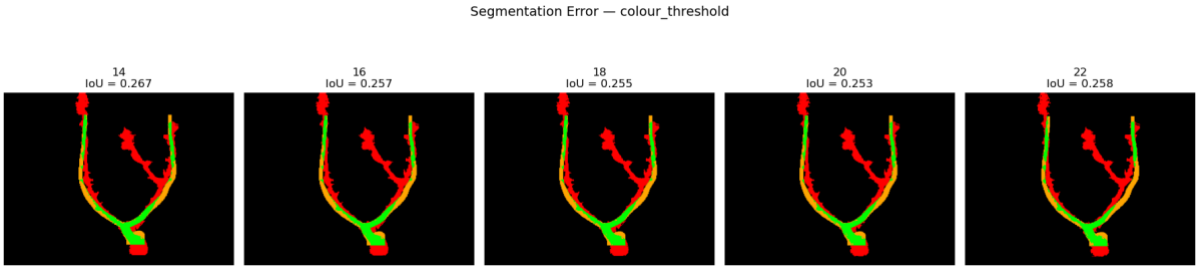
Different threshold values were tested to investigate how they affected the amount of the tree detected and the amount of unwanted background included.



Colour Threshold

The colour threshold is used to distinguish the tree from the background by comparing the blue and green channels of the winter RGB image.

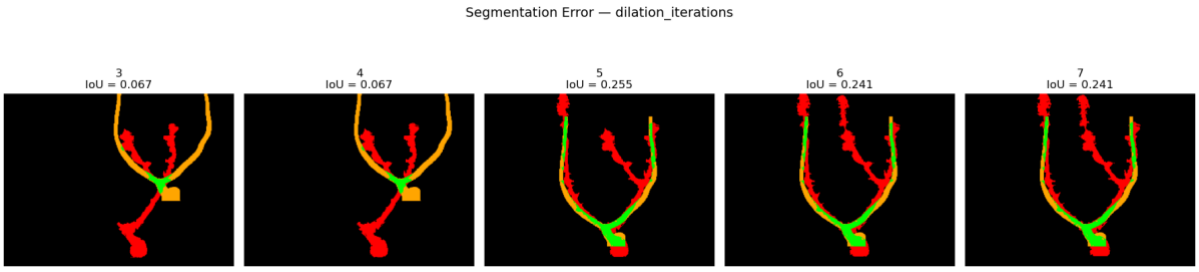
Different values were tested to investigate how changing the colour threshold affected the accuracy of the isolated tree.



Dilation Iterations

Dilation is used to expand the detected regions of the tree and help connect sections that may have been separated during the initial thresholding.

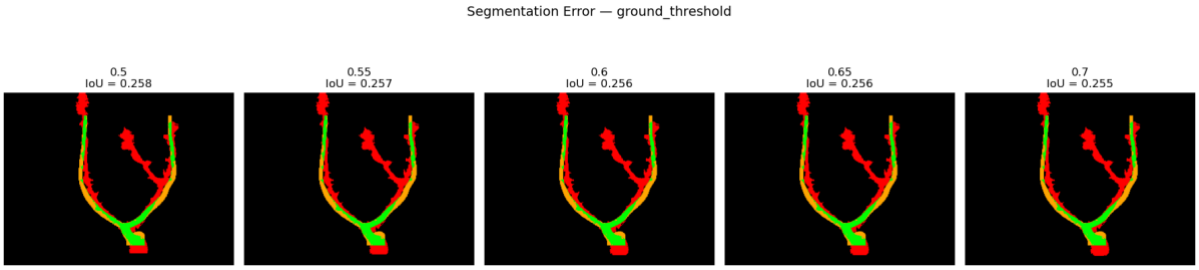
The number of dilation iterations was varied to investigate how this affected the accuracy of the segmentation.



Ground Threshold

The ground threshold determines the proportion of detected pixels required in a row for that row to be classified as part of the ground and removed.

Different ground thresholds were tested to investigate how effectively the algorithm removed the ground without removing parts of the tree.



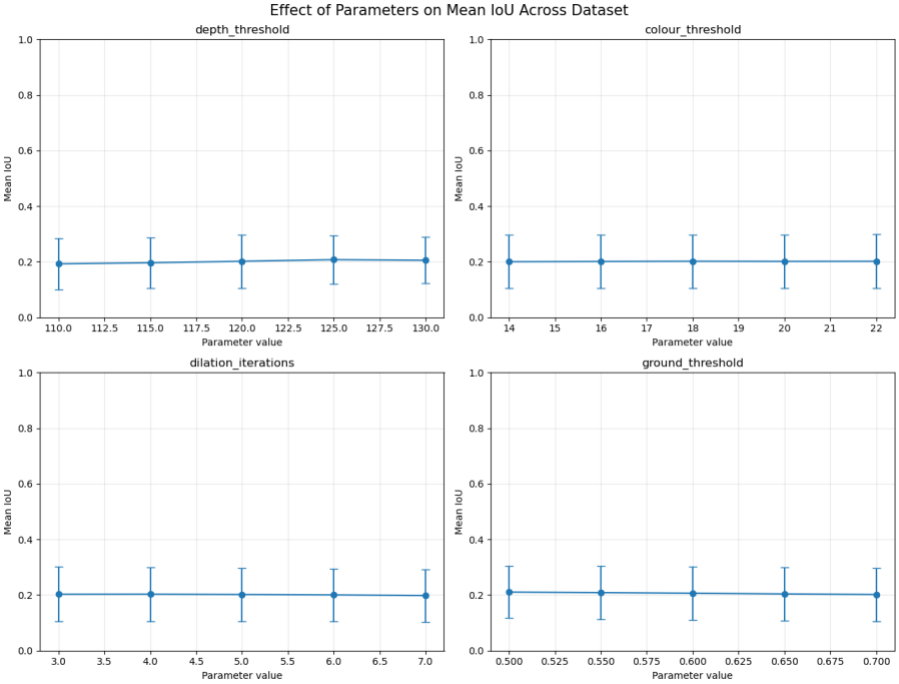
Optimisation Across Multiple Trees

After investigating the effects of the parameters on a single tree, the optimisation was extended to the full dataset of 562 trees. Each parameter was varied independently while the remaining parameters were kept at their original values.

For each parameter value, the algorithm attempted to isolate the tree, align the corresponding ground truth and calculate the Intersection over Union. The mean IoU was then calculated across the successfully evaluated observations.

The results are shown in the figure below. Although the changes in mean IoU were relatively small, the investigation allowed the parameter values to be selected quantitatively rather than relying entirely on visual inspection.

The error bars also indicate considerable variation in performance between individual trees. This suggests that no single parameter value performs equally well across every observation, and that the effectiveness of the algorithm depends partly on the characteristics of each image.



Final Parameter Selection and Comparison

Following the individual parameter investigations, the values producing the highest average IoU were selected and combined to create an optimised version of the algorithm.

The table below compares the original parameter values with the values selected through single-tree experimentation and dataset-wide optimisation.

Parameter	Original	Optimised (for 1 tree)	Optimised (data set)
Depth threshold	120	110	125
Colour Threshold	18	14	18
Dilation Iterations	5	5	4
Ground Threshold	0.7	0.5	0.5

The original and optimised configurations were then compared using the first 40 trees in the dataset. Of these, 29 trees were successfully evaluated under both configurations and included in the final paired comparison.

Evaluation Metric	Result
Original Mean IoU	0.1825
Optimised Mean IoU	0.1950
Relative Change	+6.84%

The optimised configuration increased the mean IoU from 0.1825 to 0.1950, representing a relative improvement of 6.84%. This indicates that the selected parameter changes improved the measured agreement between the predicted tree masks and the aligned ground truth for the evaluated sample.

However, the overall IoU remained relatively low. This may partly be explained by differences in viewpoint and visible tree structure between the summer ground truth and winter observations, which could reduce the measured overlap even when the tree was successfully isolated.

Another limitation was that each parameter was optimised independently rather than testing every possible combination. As a result, interactions between the parameters were not fully investigated, and the selected configuration cannot be guaranteed to be globally optimal.

Furthermore, the trees used in the final comparison were drawn from the same dataset used during optimisation. Therefore, although the results demonstrate an improvement on the evaluated sample, the optimised configuration has not been independently validated on unseen observations.

Limitations and Lessons Learned

Limitations

Ground Truth Alignment

To compare the ground truth with the isolated tree, I used a single-point alignment method with a fixed scaling factor. Although this provided an approximate alignment, it could not account for the different camera angles and distances between the summer and winter observations. This was one of the main limitations of the project because it affected the reliability of the quantitative validation.

Threshold Sensitivity

Both the colour and depth thresholds were kept constant across the dataset. This meant that a threshold which worked well for one tree might not perform as effectively for another. An improvement could involve developing adaptive thresholds based on the characteristics of each image rather than using fixed values.

Connected component assumptions

Selecting the largest connected region assumes that the tree being investigated is the largest connected object in the image. If a neighbouring tree or another object forms a larger connected region, the algorithm may select the wrong object. A possible improvement would be to introduce a centre-weighting factor to prioritise regions located near the centre of the image.

Ground Removal

The ground removal method identifies horizontal rows containing a high proportion of foreground pixels. Although this helped remove the ground, it could also remove parts of the tree if sufficiently large horizontal branches were present. A possible improvement would be to consider the shape and orientation of detected structures rather than relying only on the proportion of foreground pixels in each row.

Unusual images

The algorithm relies on several assumptions about the appearance and structure of the trees. Unexpected objects, unusual lighting conditions or differences in tree shape could reduce its accuracy. More advanced image-processing methods or machine-learning approaches could potentially improve its ability to handle these variations.

Lessons Learned

Combining Different Methods

While depth thresholding alone was insufficient to isolate the tree reliably, combining it with colour segmentation, morphological dilation, ground removal and connected component analysis produced a more effective result. This demonstrated the importance of combining complementary computational methods when a single approach is insufficient.

Importance of Validation

Although ground-truth data was provided, differences between the summer and winter observations meant that a direct comparison was not possible. This demonstrated that the reliability of an evaluation metric depends not only on the algorithm being tested but also on the quality and suitability of the reference data.

Numerical Optimisation

The project highlighted the importance of quantitative optimisation rather than relying entirely on visual inspection. By comparing different parameter values using Intersection over Union, it was possible to investigate how changes to the algorithm affected its measured performance and identify improvements.

Conclusion

This project developed a Python-based algorithm to isolate tree structures from winter RGB and depth images. By combining several image-processing techniques, the algorithm was able to produce binary masks identifying the main tree.

The initial evaluation achieved a mean IoU of 0.2019 across 422 successfully evaluated trees. Subsequent parameter optimisation demonstrated that the measured performance could be improved, with the paired comparison showing a 6.84% relative increase in mean IoU.

The main limitation was the difference between the summer ground truth and winter observations, which prevented a fully reliable pixel-by-pixel comparison. Nevertheless, the project demonstrated how image-processing methods, quantitative validation and parameter optimisation can be combined to develop and evaluate a computational solution to an engineering problem.